Notes on Introductory Koopman Operator Theory

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Contents

1 Mathematical Preliminary Notions **5**

2 Introduction to Koopman Operator Theory **9**

2.1 Koopman Operator in Discerete Time 9

2.1.1 Example: Rotation on a Circle: Pure Point Spectrum 11

2.2 Continuous-time Setting: Koopman Generator 11

3 Koopman Invariant Subspaces & DMD **15**

3.1 Modal Analysis Notion : Linear Systems 16

3.2 Review of the Koopman operator 17

3.3 Introduction to Koopman Modes 18

3.3.1 Koopman Modes for a Linear System 19

3.3.2 Koopman Invariant Subspaces & Finite-Dimensional Linearization 21

3.4 Introduction to Dynamic Mode Decomposition (DMD) 23

3.4.1 DMD with leading spectral decomposition 24

4 Bilinear Model Learning **27**

4.1 Bilinear Models Approximating Koopman Generator 27

4.1.1 Lifted LTI Model vs Bilinear Model 28

4.2 Approximation of the Koopman Generator using HMM 28

5 Appendix **31**

5.1 MATLAB Codes 31

Chapter 1

Mathematical Preliminary Notions

Some introductory mathematical notions are provided here.

Dynamical Systems: Mathematically, (X, F) is a dynamical system where,

- X is the space (state space), usually a compact metric.
- F is the evolution rule, usually a map from X to itself. We study the orbit of points $\mathbf{x} \in X$. Where, $\mathbf{x}$ evolve with time as per F. So we say, $\dot{\mathbf{x}}(t) = F(\mathbf{x}(t))$ or in a discrete-time setting, $\mathbf{x}_{k+1} = F(\mathbf{x}_k)$. Often, $\mathbf{x}$ doesn't depend explicitly on time, so we simply say, $\dot{\mathbf{x}} = F(\mathbf{x})$.
- The *orbit/trajectory* of $\mathbf{x}$ is a time-ordered set of states that can be defined as,

$$orb(\mathbf{x}) = x, F(x), \underbrace{F \circ F(x)}_{F^2(x)}, \underbrace{F \circ F \circ F(x)}_{F^3(x)}, ...$$

Ergodic theory is the branch of mathematics that studies statistical properties of deterministic dynamic systems by means of invariant measures.

Invariant sets: Consider an autonomous and time-invariant dynamical system represented as $\dot{x} = f(x)$ with $f : \mathbb{R}^n \to \mathbb{R}^n$. A set $\mathcal{A} \subseteq \mathbb{R}^n$ is said to be invariant (w.r.t. the system or f) if for every trajectory x, $x(t) \in \mathcal{A} \implies x(\tau) \in \mathcal{A}$ for all $\tau \geq t$. It means that a trajectory entering $\mathcal{A}$ or starting in $\mathcal{A}$ stays in $\mathcal{A}$.

Closed set (under operation): A set $\mathcal{A}$ is closed if under an operation on the members of $\mathcal{A}$ always produces a set $\mathcal{D} \subseteq \mathcal{A}$. This means that performing an operation on any two numbers in $\mathcal{A}$ always results in a number belonging to $\mathcal{A}$. Examples of closed sets: $\mathcal{R}$ under addition and $\mathcal{R}$ under multiplication.

$\boldsymbol{\sigma}$-algebra on a set $\mathcal{X}$ is a non-empty collection Σ of subsets of $\mathcal{X}$ which is closed under *complement, finite unions,* and *finite intersections.* The ordered pair $(\mathcal{X}, \Sigma)$ is called a **measurable space**.

Measure space: A measure is a ordered triple $(\mathcal{X}, \Sigma, \mu)$ consisting of,

- $\mathcal{X}$ is a set.
- Σ is a σ-algebra on $\mathcal{X}$.

- μ is a measure on $(\mathcal{X}, \Sigma)$.

Example: Let us consider the set $\mathcal{X} = \{0, 1\}$. The power set (the set of all subsets) Σ is a σ-algebra on $\mathcal{X}$. Therefore, $\Sigma = \{\phi, \{0\}, \{1\}, \{0, 1\}\}$. The measure μ is defined as $\mu(\{0\}) = \mu(\{1\}) = \frac{1}{2}, \mu(\mathcal{X} = 1)$, and $\mu(\phi) = 0$.

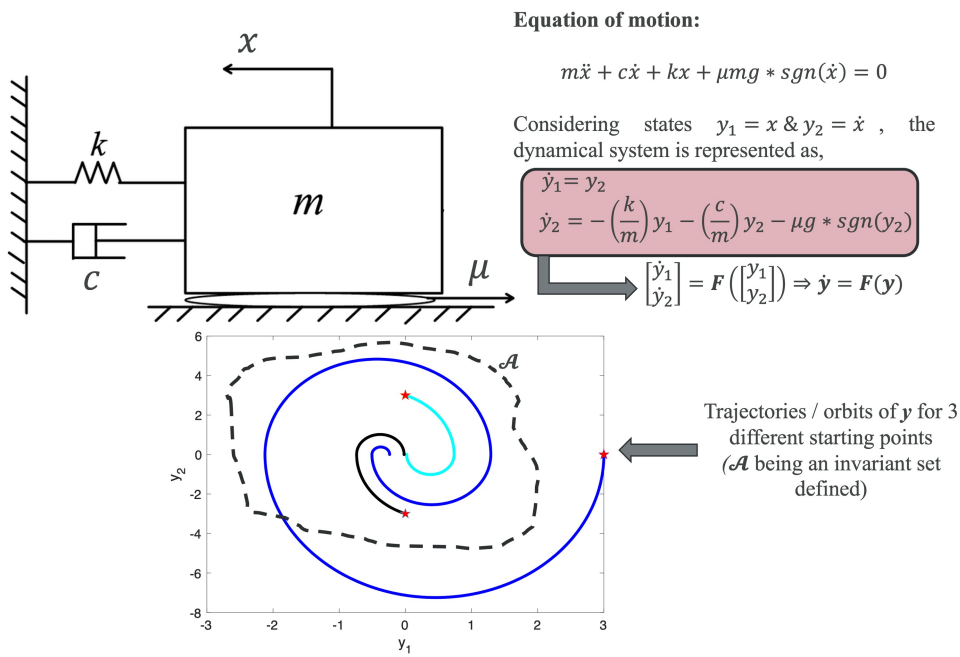

Figure 1.1: An example nonlinear dynamical system with its orbit and an invariant set shown

Measurable function: A measurable function is a function that operates between the sets of two measurable spaces and maintains the structure of those spaces. Let, $(\mathcal{X}, \Sigma_x)$ and $(\mathcal{Y}, \Sigma_y)$ are two measurable spaces. A function $f : \mathcal{X} \to \mathcal{Y}$ is measurbale if for every $E \in \Sigma_y$, the pre-image $f^{-1}(E) \in \Sigma_x$.

Square integrable function: A function $f : \mathbb{R} \to \mathbb{C}$ is square integrable $\iff \int_{-\infty}^{\infty} f(x)dx < \infty$. A similar definition can be drawn for a function with its domain $\mathcal{D} \subseteq \mathbb{R}$. In measure theory, for a measure space $(\mathcal{X}, \Sigma, \mu)$, square integrable functions are denoted as $L^2(X, \mu)$ or $L^2(X)$ or L^2 that satisfy $\int_{\mathcal{X}} |f(x)|^2 d\mu(x) < \infty$.

Normed vector space: It consists of an underlying vector space V over a field of scalars (real or complex), together with a norm $||.|| : V \to \mathbb{C}$ or $\mathbb{R}^+$ that satisfies,

- $||v|| = 0 \iff v = 0$.
- $||\alpha v|| = |\alpha|||v||$ whenever α is a scalar .
- $||v + w|| \leq ||v|| + ||w||$ for all $v, w \in V$

Inner product space: It is a vector space V over a field of scalars (real or complex), together with an inner product $<,>: V \cdot V \to \mathbb{R}$ or $\mathbb{C}$, that satisfies,

- $< u, v >= \overline{< v, u >}$ for all $u, v \in V$

- Linearity in the first argument, i.e., $< \alpha u + \beta v, w >= \alpha < u, w > + \beta < v, w >$.
- Positive definiteness, i.e., $< v, v > \ \geq 0$. With, $< v, v >= 0 \iff v = 0$.

Banach Space: A normed vector space that is complete. ($** \ L^2$ or the family of L^p are Banach spaces).

Hilbert space: An inner product space that is complete. (** L^2 is the only Hilbert space among L^p family of spaces).

Birkhoff's Ergodic Theorem: Let an ergodic dynamical system be (X, F) & and a map $g : X \to \mathbb{C}$ is a measurable function with measure μ. Then, for all $\mathbf{x} \in X$, space average = time average, i.e.,

$$\lim_{n\to\infty} \frac{1}{n}[g(x) + g(F(x)) + g(F^2(x)) + ... + g(F^{n-1}(x))] = \int_X g \ d\mu$$

$$\implies \lim_{n\to\infty} \frac{1}{n} \sum_{i=0}^{n-1} g \circ F^i(\mathbf{x}) = \int_X g \ d\mu$$

Isometry: Let (X, d_x) and (Y, d_y) be two metric spaces. A map $f : X \to Y$ is called isometric if for any $a, b \in X$ we have $d_x(a, b) = d_y(f(a), f(b))$.

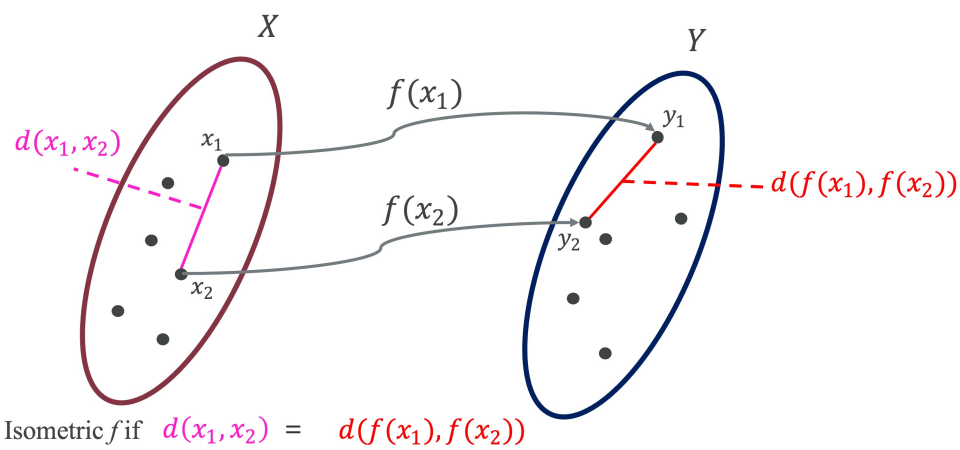

Figure 1.2: Graphical depiction of isometry

Adjoint of an operator: Consider a linear operator $L : \mathcal{V} \to \mathcal{V}$,, where $\mathcal{V}$ is an inner product space. The adjoint of L is an operator L^* satisfying $< L\mathbf{x}, \mathbf{y} >=< \mathbf{x}, L^*\mathbf{y} >$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{V}$.

Unitary Operator: A unitary operator is a surjective bounded operator on a Hilbert space that preserves the inner product. A unitary operator satisfies $L^*L = LL^* = I$, where I is the identity operator.

Group Homomorphism: It is a function $g : G \to H$ such that for all $u, v \in G$ satisfy $h(u * v) = g(u)g(v)$. A continuous group homomorphism is termed a **one-parameter group**.

A Strongly continuous one-parameter unitary group (U^t):

- *Strong continuity:* $\lim_{s\to t} ||U^t f - U^s f|| = 0 \; \forall f \in \mathcal{H}$ and $s, t \in \mathbb{R}$.
- *Group property:* $U^{t+s} = U^t U^s = U^s U^t$ and $U^0 = I$ $(\forall s, t \in \mathbb{R})$.
- *Unitary property:* $U^{t*} = U^{-t}$ $(\forall t \in \mathbb{R})$

Theorem 1.0.1. ***Stone's theorem:*** *If U^t is a strongly continuous one-parameter unitary group, there exists a unique self-adjoint operator $V : \mathcal{D}_V \to \mathcal{H}$ with $D_V \subseteq \mathcal{H}$ such that,*

$$U^t = e^{itV} = \sum_{n=0}^{\infty} \frac{(it)^n V^n}{n!} \; (\forall t \in \mathbb{R}) \tag{1.0.1}$$

with the domain of V defined as,

$$\mathcal{D}_V = \{f \in \mathcal{H} : \lim_{t\to\infty} \frac{1}{t}(U^t f - f) \text{ exists}\}.$$

The operator V is called ***infinitesimal generator*** *of U^t and can be computed as,*

$$Vf = \lim_{t\to 0} \frac{1}{it}(U^t f - f) \tag{1.0.2}$$

Chapter 2

Introduction to Koopman Operator Theory

The modeling of dynamical systems involves finding ways to describe the evolution of the system's state. In ergodic theory, dynamic systems are described as the evolution of functions in the state space. The evolution of functions is governed by a linear operator called the Koopman operator.

Ergodic theory studies the steady-state behavior of systems. In steady-state, for many instances, a system's state may become an invariant set (for example, periodic orbit and chaotic attractor). In such cases, we can define a probability measure reflecting the likelihood of finding the state in a given subset of the state space. If there exists such a probability measure that does not change with time (time-invariant), then the system is called as measure preserving. In this case, the system has a natural functional space, for example, the space of functions that are square integrable w.r.t. the probability measures. In such a functional space, the Koopman operator acts isometrically (preserves the norm). The spectrum of the Koopman operator can tell us valuable information about the long-term behavior of the system.

In practical control applications, we are not only interested in the steady-state behavior but also in the transient behavior. It may seem that the analysis provided by ergodic theory is not instrumental. However, operator-based methods may also be applied to study transient behavior along with steady-state behavior. There may be some loss of structure in such cases. For example, the Koopman operator may lose the isometric property, but its spectrum is still capable of providing valuable information about a system's transient behavior.

Koopman operator, being linear, enables us to do modal analysis for linear systems to be extended to non-linear systems.

2.1 Koopman Operator in Discerete Time

Consider a discrete-time dynamical system with a state-variable x belonging to a smooth manifold $\mathcal{M}$ with a map $F : \mathcal{M} \to \mathcal{M}$ given as,

$$x_{k+1} = F(x_k) \tag{2.1.1}$$

Here, k is an integer index representing the index of time, and $\mathcal{M}$ is a measure-space that can be expressed as $(\mathcal{M}, \Sigma, \mu)$. Also, we consider that the F is measure preserving, i.e.,

$$\mu(F^{-1}A) = \mu(A) \ \forall A \in \Sigma \tag{2.1.2}$$

Let us consider an observable function of the states in a space $\mathcal{F}$ (where $\mathcal{F}$ is the space of square-integrable functions $\mathcal{L}^2(\mathcal{M}, \mu)$, or in general a *Banach* space $\mathcal{B}$/ *Hilbert* space $\mathcal{H}$). For, $g \in \mathcal{F}$, the Koopman operator $U : \mathcal{F} \to \mathcal{F}$ is defined as,

$$\begin{aligned} Ug(x_k) = g \circ F(x_k) \text{ or } g(F(x_k)) \\ \implies Ug(x_k) = g(x_{k+1}) \end{aligned} \tag{2.1.3}$$

If F is measure preserving, then the Koopman operator is an isometry, i.e.,

$$\begin{aligned} ||Ug(x_k)||^2 = ||g(F(x_k)||^2 \\ \text{or in a general sense, } ||Ug||^2 = ||g||^2 \end{aligned} \tag{2.1.4}$$

Proof: The Koopman operator is an isometry

$$\begin{aligned} ||Ug(x_k)||^2 &= \int_{\mathcal{M}} |g(F(x_k)|^2 d\mu(x_k) \\ &= \int_{\mathcal{M}} |g(x_{k+1})|^2 d\mu(F^{-1}(x_{k+1})) \\ &= \int_{\mathcal{M}} |g(x_{k+1})|^2 d\mu(x_{k+1}) \\ &= ||g(x_{k+1}||^2 \\ &= ||g(F(x_k)||^2 \end{aligned}$$

The Koopman operator, being a linear one, is unitary if F is invertible. Otherwise, it just acts on g isometrically.

Spectral properties of the Koopman operator can reveal important information about the dynamical system under study. Let Φ be an eigenfunction of U with eigenvalue λ. Therefore,

$$U\Phi = \lambda\Phi \tag{2.1.5}$$

From 2.1.5, we can write,

$$||U\Phi||^2 = |\lambda|^2||\Phi||^2 \tag{2.1.6}$$

As the Koopman operator is an isometry due to the measure-preserving nature of F, we can write using 2.1.4 and 2.1.6,

$$||U\Phi||^2 = |\lambda|^2||\Phi||^2 = ||\Phi||^2 \tag{2.1.7}$$

From 2.1.7 it is clear that $|\lambda| = 1$ and therefore, with U an isometry, $\lambda = e^{i\theta}$. So, for a measure-preserving system, the magnitudes of the eigenvalues of the Koopman operator are unity, and they lie on the unit circle.

Using the spectral information of the Koopman operator, we can define an observable variable or coordinate as,

$$z_k = \Phi(x_k) \text{ or } z_{k+1} = \Phi(x_{k+1}) \tag{2.1.8}$$

Now, using 2.1.3 and 2.1.5,

$$z_{k+1} = U\Phi(x_k) \implies z_{k+1} = \lambda\Phi(x_k) \implies z_{k+1} = \lambda z_k \tag{2.1.9}$$

Therefore, the evolution of the observable variable z is linear. For a measure-preserving system, $|\lambda| = 1$, and therefore, the magnitude of z is constant along a trajectory, and its dynamics are simply rotations by an angle θ.

Ergodic system: A measure-preserving system is said to be ergodic if the only eigenfunctions Φ that satisfy $U\Phi = \Phi$ are constant almost everywhere.

2.1.1 Example: Rotation on a Circle: Pure Point Spectrum

Consider the state space $\mathcal{M} = [0, 1)$ with Lebesgue measure, and the dynamics given by,

$$F(x) = x + \alpha \tag{2.1.10}$$

Consider the Koopman operator U in the Banach Space/Hilbert Space $\mathcal{F}$. Consider the function $\Phi_n \in \mathcal{F}$ given as $\Phi_n(\tau) = e^{2\pi i n\tau}$ $(n \in \mathbb{Z})$. Now,

$$\begin{aligned} U\Phi_n(x) = \Phi_n(F(x)) = \Phi_n(x+\alpha) = e^{2\pi i n(x+\alpha)} = e^{2\pi i n\alpha}e^{2\pi i n x} = e^{2\pi i n\alpha}\Phi_n(x) \\ \implies U\Phi_n(x) = \lambda_n\Phi_n(x) \end{aligned} \tag{2.1.11}$$

where, $\lambda_n = e^{2\pi i n\alpha}$

Therefore, from 2.1.11, clearly Φ_n is an eigenfunction of U with eigenvalue λ_n. The eigenfunctions form an orthonormal basis, and the spectrum of U lies on the unit circle. Therefore, U has a pure point spectrum. Consider two scenarios:

- α **is rational:** for example, $\alpha = 1/4$, then the eigenvalues are ± 1 and $\pm i$ $(\forall n \in \mathbb{Z})$.For the eigenvalue $+1$, the corresponding eigenfunctions are $\Phi_0, \Phi_{\pm 4}, \Phi_{\pm 8}$..... Therefore, $\Phi_0 = 1$ is not the only eigenfunction (or only constant eigenfunction). Therefore, the system is not ergodic.
- α **is irrational:** Then there are a countably infinite number of eigenvalues densely found on the unit circle. In this case, Φ_0 is the only eigenfunction corresponding to the eigenvalue $+1$. Therefore, the system is ergodic.

For this case, λ_1 and Φ_1 are a special pair of eigenvalue and eigenvector as all others can be determined from them, i.e., $\lambda_n = (\lambda_1)^n$ and $\Phi_n = (\Phi_1)^n$. There will be eigenvalues arbitrarily close to λ_1 with eigenfunctions arbitrarily highly oscillatory. To distinguish between the eigenvalues and their corresponding eigenfunctions, with norms of all eigenfunctions to be unity, we order them as per their Dirichlet energy (Ginnakis 2019 [3]) given as,

$$Energy(\Phi) = ||\nabla\Phi||^2 \tag{2.1.12}$$

For this example, $Energy(\Phi_n) = ||\partial\Phi_n/\partial x||^2 = (2\pi n)^2$

2.2 Continuous-time Setting: Koopman Generator

Let us consider a map $F : \mathcal{M} \to \mathcal{M}$, on a smooth manifold $\mathcal{M}$, that governs the dynamics of a system with state $x(t) \in \mathcal{M}$ $(t \in \mathbb{R})$ as

$$\dot{x} = F(x) \tag{2.2.1}$$

Let the flow of the system in 2.2.1 is $\Psi(t,x) : \mathbb{R} \times \mathcal{M} \to \mathcal{M}$, represented as $\Psi^t(x)$.

$$\begin{aligned} x(t) = \Psi(t,x) = \Psi^t(x) \text{ with } (\Psi^0(x) = x) \\ \frac{d\Psi^t(x)}{dt} = F(x) \end{aligned} \tag{2.2.2}$$

As in the discrete-time case, we consider that $(\mathcal{M}, \Sigma, \mu)$ a measure space and Ψ^t is measure-preserving. Similarly, for an observable function g in $\mathcal{L}^2(\mathcal{M}, \mu)$ space $\mathcal{F}$ (or Banach / Hilbert space), we define the continuous-time Koopman operator U^t as

$$U^t g = g \circ \Psi^t \tag{2.2.3}$$

Considering a time-varying observable $h(t,x)$ with $h(0,x) = g(x)$, we can equivalently express 2.2.3 as,

$$U^t g(x) = h(t,x) \tag{2.2.4}$$

It can be shown that the time-varying observable function h satisfies,

$$\frac{\partial}{\partial t} h(t,x) = L_F g(x) \tag{2.2.5}$$

Where L_F is the Lie derivative w.r.t. F given as,

$$L_F = \lim_{t \to 0} \frac{U^t - I}{t} \tag{2.2.6}$$

L_F is said to be the *infinitesimal generator* of the Koopman operator.

Proof: $\frac{\partial}{\partial t} h(t,x) = L_F g(x)$

$$\begin{aligned} \frac{\partial}{\partial t} h(t,x) &= \frac{\partial}{\partial t} U^t g(x) \\ &= \frac{\partial}{\partial t} g \circ \Psi^t(x) \\ &= \frac{\partial g}{\partial \Psi^t} . \frac{\partial \Psi^t}{\partial t} \\ &= \nabla g(x).F(x) \\ &= (F.\nabla) g(x) \\ &= L_F g(x) \end{aligned}$$

Where, $L_F = (F.\nabla)$

Proof: $L_F = \lim_{t \to 0} \frac{U^t - I}{t}$ and L_F is the infinitesimal generator of the Koopman operator.

For each time t, we obtain the respective Koopman operator $U^t : \mathcal{F} \to \mathcal{F}$. For an invertible Ψ^t, U^t satisfies the group property and is also unitary. Therefore, U^t qualifies to be a one-parameter unitary group. Therefore, there exists an infinitesimal generator V of the Koopman operator and can be defined as (using Stone's theorem 1.0.1),

$$U^t = e^{tV} \tag{2.2.7}$$

$$Vg = \lim_{t \to 0} \frac{U^t g - g}{t} \tag{2.2.8}$$

2.2.8 can be equivalently written as,

$$Vg = \lim_{t \to 0} \frac{U^t - I}{t} g(x) \tag{2.2.9}$$

Again, 2.2.8 $\implies$

$$\begin{aligned} Vg &= \lim_{t \to 0} \frac{g \circ \Psi^t(x) - g(x)}{t} \\ &= \lim_{t \to 0} \frac{g(\Psi^t(x)) - g(\Psi^0(x))}{t} \\ &= \nabla g . \lim_{t \to 0} \frac{\Psi^t(x) - \Psi^0(x)}{t} \\ &= \nabla g . \frac{d\Psi^t(x)}{dt} \\ &= \nabla g . F(x) \\ &= (F.\nabla) g \\ \therefore Vg &= L_F g \end{aligned} \tag{2.2.10}$$

Therefore, L_F is the infinitesimal generator of the Koopman operator.

Comparing 2.2.9 & 2.2.10 gives, $L_F = \lim_{t \to 0} \frac{U^t - I}{t}$.

From 2.2.7, it is clear that U^t and L_F have the same eigenvector Φ with the following equivalency:

$$L_F \Phi = \lambda \Phi \iff U^t \Phi = e^{\lambda t} \Phi \tag{2.2.11}$$

With a time-varying eigenfunction $\varphi(t, x)$ with $\varphi(0, t) = \Phi(x)$, we can define an observable $z(t) = \Phi(x)$ such that,

$$\begin{aligned} \dot{z}(t) = \frac{\partial}{\partial t} \varphi(t, x) = L_F \Phi(x) = \lambda \Phi(x) \\ \implies \dot{z}(t) = \lambda z(t) \end{aligned} \tag{2.2.12}$$

In this way, the coordinate z gets decoupled from others and evolves linearly with time.

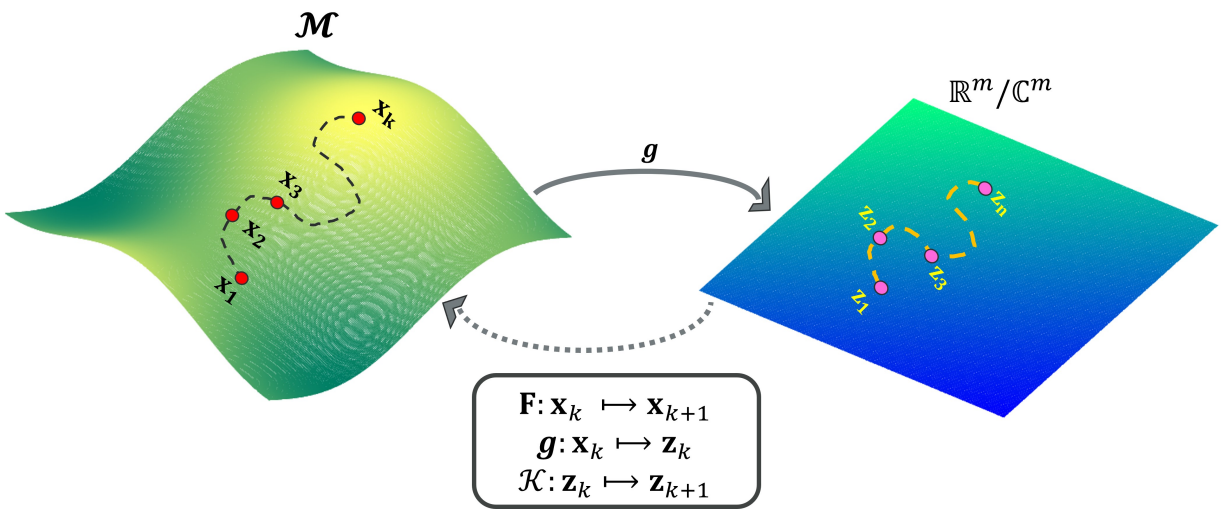

Figure 2.1: Schematic of Koopman operator for nonlinear systems (adapted from [4])

Chapter 3

Koopman Invariant Subspaces & DMD

Koopman operator describes the evolution of a dynamical system in time in terms of Hilbert space functions. These functions, referred to as observables, can be outputs of a n experiment of simulation. The Koopman operator being linear enables the notion of modal analysis to be extended to non-linear systems. Since Koopman operator is infinite dimensional, we are interested in finding an invariant subspace under the operation of it so we can have a finite-dimensional approximation. For example, the level sets of Koopman eigenfunctions can form invariant subspaces under which the dynamical system behave in ergodic manner. Projection of the observables onto the eigenfunctions of the Koopman operator decomposes the observables into a linear superposition of Koopman modes with their coordinates/coefficients given by the Koopman eigenfunctions.

The DMD algorithm provides an approximation of the Koopman operator and its modes by considering the full state as observable. The EDMD algorithm approximates the Koopman operator on a dictionary of observable functions from data. If the span of the observables included in the dictionary contains a Koopman invariant subspace, then with enough data, the representation of the dynamical system in this subspace will be exact. EDMD is a Galerkin method, withn a data-driven inner product as long as enough data is used (i.e., $rank(data) \approx dim(\mathcal{F}_s)$, where $\mathcal{F}_s$ is the subspace spanned by the observables). However, the choice of dictionary is ad-hoc without any clear guidelines to achieve the span of a Koopman invariant subspace.

One approach to mitigate the cost of a large dictionary in EDMD is to formulate it as a Kernel DMD. However, there is still no clear guidelines on the selection of kernel functions. Otto et al. [2] proposed a new technique for learning of the observable functions efficiently (i.e., with a very small set of informative observables) from data that span a Koopman invariant subspace. The key contributions of [2] can be highlighted below:

- Two neural-networks in a architecture similar to an undercomplete autoencoder are proposed representing the collection of observables together with a non-linear reconstruction of full-state from these features.
- A learned linear transformation transforms the observables in time as in an RNN, providing the finite-dimensional approximation of the Koopman operator $\mathbf{K}_{approx}$ on the subspace of the observables.
- This approach is different from the previously existing neural-networks to learn dictionaries

for DMD because it uses a second network to reconstruct the full state.

- Previuosly presented methods use linear state recosntruction using the Koopman modes assuming that the full state observable of the system lies in the leraned Koopman invariant subspace. The method presented in this study uses non-linear reconstruction which relaxes this assumption.
- The method minimizes the state reconstruction error over several time steps into the future, providing the possibility to detect highly observable features even with small amplitudes similar to Visual Interaction Networks (VINs) to learn physics-based dynamics models for encoded latent variables.

3.1 Modal Analysis Notion : Linear Systems

Modal analysis of dynamical systems or systems of ODEs is the process of finding the linear choice of coordinates where the ODEs of the system are decoupled from one another. The new coordinates are called modal coordinates. Often, dynamical systems involve a system of coupled ODEs, which may sometimes be complex to analyze or understand. A set of decoupled ODEs that are representatives of the coupled ODEs provides us with easier ways to understand the system behavior and solve the dynamics. The behavior of the system in the modal coordinates can be translated back to the original coordinates. The modal analysis also provides the system behavior in terms of the individual behavior of the modes.

Example: Consider a mass-spring-damper system represented by the following differential equation,

$$m\ddot{x} + c + kx = 0$$

Considering $\mathbf{y} = \begin{bmatrix} y_1 & y_2 \end{bmatrix}^T = \begin{bmatrix} x & \dot{x} \end{bmatrix}^T$, the equation of motion can be expressed in the state-space form of a system of coupled ODEs as,

$$\underbrace{\begin{bmatrix} \dot{y_1} \\ \dot{y_2} \end{bmatrix}}_{\dot{\mathbf{y}}} = \underbrace{\begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}}_{\mathbf{A}} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Now we consider the modal co-ordinate be $\mathbf{z} = \begin{bmatrix} z_1 & z_2 \end{bmatrix}^T$, such that,

$$\mathbf{y} = \mathbf{T}\mathbf{z}$$

Here $\mathbf{T}$ is a modal transformation matrix. $\mathbf{T}$ is the matrix comprising the orthonormal eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ associated with the eigenvalues λ_1 and λ_2 respectively.

Now, we multiply using the transformation and write the system of ODEs as,

$$\mathbf{T}\dot{\mathbf{z}} = \mathbf{A}\mathbf{T}\mathbf{z} \implies \dot{\mathbf{z}} = \mathbf{T}^{-1}\mathbf{A}\mathbf{T}\mathbf{z} \implies \begin{bmatrix} \dot{z_1} \\ \dot{z_2} \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

Therefore, we have two decoupled ODEs in the modal coordinates, which can be solved easily and obtain a solution. Once we obtain $\mathbf{z}$, we can obtain the solution in original coordinates using the relation $\mathbf{y} = \mathbf{T}\mathbf{z}$. Consider an example with $m = 1, k = 20$, and $c = 12$, and the initial conditions $x(0) = 0.2, \dot{x}(0) = 0.5$ as solutions shown in figure 3.1.

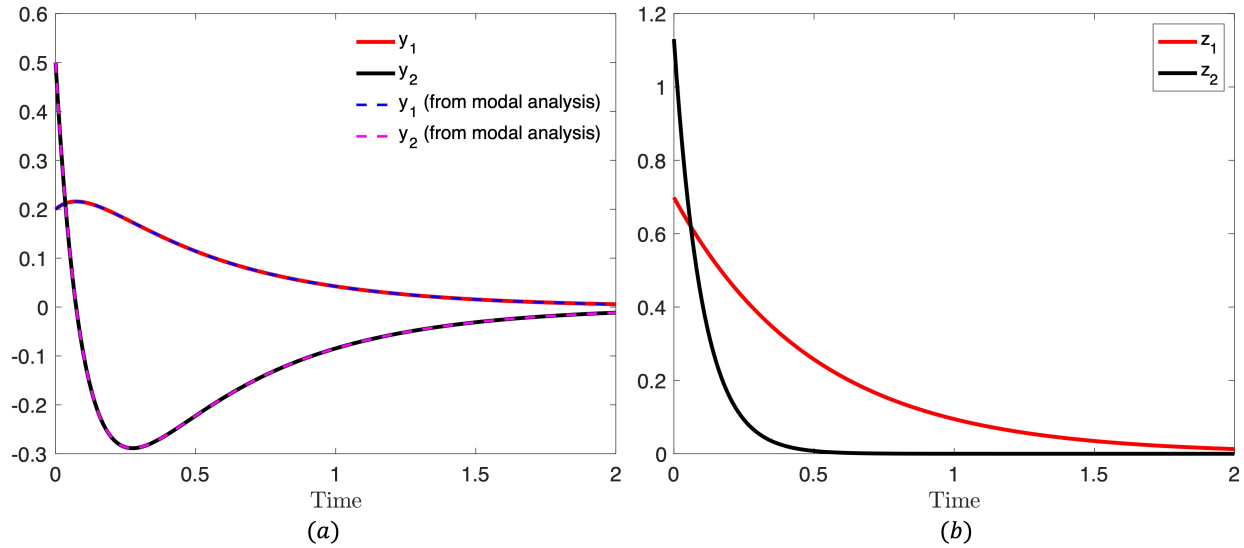

Figure 3.1: (a) Comparison of the solutions of the coupled ODE and solutions obtained from modal analysis (from the decoupled ODEs of the modal coordinates), (b) The evolution of the states in the modal coordinates

3.2 Review of the Koopman operator

Consider a manifold $\mathcal{M} \subseteq \mathbb{R}^n$ be the state-space of a dynamical system and $\mathbf{f} : \mathcal{M} \to \mathcal{M}$ be its vector field so that the dynamics is given as,

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) \tag{3.2.1}$$

The flow of the state varibale with $\mathbf{x}(t_0) = \mathbf{x}_0$ can be defined as,

$$\mathbf{F}^t(\mathbf{x}_0) = \mathbf{x}(t + t_0) = \mathbf{x}_0 + \int_{t_0}^{t+t_0} f(\mathbf{x}(\tau))d\tau \tag{3.2.2}$$

If we take discrete measurements in the time interval $[t_0, t]$ as $\{t_0, t_1, t_2, \ldots, t_k\}$ with sampling time $\Delta t = t_i - t_{i-1}$ so that the discrete time evolution of the state-vector is given as $\{\mathbf{x}_{t_0}, \mathbf{x}_{t_1}, \ldots, \mathbf{x}_{t_k}\} := \{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_2 \ldots, \mathbf{x}_k\}$ (note that $\mathbf{x}_1 = \mathbf{x}_{t_0} = \mathbf{x}_0$). From 3.2.2, we can write the discrete-time dynamics as,

$$\mathbf{x}_{k+1} = \mathbf{F}^k(\mathbf{x}_1) = \mathbf{F}^{k-1}(\mathbf{x}_2) = \cdots = \mathbf{F}^{k-(k-1)}(\mathbf{x}_k)$$

$$\implies \mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k) \tag{3.2.3}$$

Let $\mathcal{F} \subseteq \mathcal{H}$ be the space of complex valued scalar observable function $g : \mathcal{M} \to \mathbb{C}$ ($g \in \mathcal{F}$). The Koopman operator $\mathcal{K}$ acts on the observable as,

$$\mathcal{K}g = g \circ \mathbf{F} = g(\mathbf{F}(\mathbf{x}_k)) \tag{3.2.4}$$

$$\implies \mathcal{K}g(\mathbf{x}_k) = g(\mathbf{x}_{k+1}) \tag{3.2.5}$$

Therefore, the Koopman operator describes the evolution of the observable functions on a subspace of the Hilbert space. In other words, it descibes how the states of the dynamical system evolve in terms of Hilbert space functions. These observable functions can be outputs of an experiment or simulation.

Using 3.2.5, we can write,

$$g(\mathbf{x}_k) = \mathcal{K}g(\mathbf{x}_{k-1}) = \mathcal{K}(\mathcal{K}g(\mathbf{x}_{k-2})) = \cdots = \mathcal{K}^k g(\mathbf{x}_0) \tag{3.2.6}$$

Here, $\mathcal{K}^k$ is the Koopman operator in time step k.

The subspace $\mathcal{F}$ on which the Koopman operator acts on the observable must be invariant, i.e., for $g \in \mathcal{F}$ we must also have $g \circ \mathbf{F} \in \mathcal{F}$. Care should be taken in the choice of $\mathcal{F}$. In ergodic theory, we assume that $\mathcal{M}$ is a measure space and $\mathbf{F}$ is measure-preserving which leads to easier selection of $\mathcal{F} = \mathcal{L}^2(\mathcal{M})$ and $\mathcal{K}$ acts on g isometrically.

The Koopman operator $\mathcal{K}$ is linear as it satisfies the superposition principle,

$$\mathcal{K}(\alpha g + \beta h)(\mathbf{x}_k) = \alpha\mathcal{K}g(\mathbf{x}_k) + \beta\mathcal{K}h(\mathbf{x}_k) \text{ for any } g, h \in \mathcal{F} \text{ and } \alpha, \beta \text{ scalars} \tag{3.2.7}$$

It is important to note that the dynamical system is non-linear on the finite dimensional manifold $\mathcal{M}$. On the other hand, the Koopman operator $\mathcal{K}$ is linear on the infinite dimensional space $\mathcal{F}$.

3.3 Introduction to Koopman Modes

Koopman modes provide useful coherent structures for studying a dynamical system's evolution and dominant pattern forming behaviors. Consider a dynamical system evolving on a manifold $\mathcal{M}$ as in 3.2.3. We can consider the continuous time equation as in 3.2.1. However, as we are dealing with discrete time data, we adopt the discrete-time setting.

Since the Koopman operator is linear, it may have eigenvalues and eigenfunctions. The eigenfunctions may lie in the infinite dimensional subspace $\mathcal{F}$ of the Hilbert space. Let, the eigenfunction of the Koopman operator be $\{\varphi_j\}_{j=1}^{\infty} := \{\varphi_1, \varphi_2, \dots\}$ and the corresponding eigenvalues be $\{\lambda_j\}_{j=1}^{\infty} := \{\lambda_1, \lambda_2, \dots\}$. Therefore, for any eigen pair we can write,

$$\mathcal{K}\varphi_j(\mathbf{x}_k) = \varphi_j(\mathbf{F}(\mathbf{x}_k)) = \varphi_j(\mathbf{x}_{k+1}) \tag{3.3.1}$$

and,

$$\mathcal{K}\varphi_j(\mathbf{x}_k) = \lambda_j\varphi_j(\mathbf{x}_k) \tag{3.3.2}$$

Therefore, from 3.3.1 and 3.3.2,

$$\varphi_j(\mathbf{x}_{k+1}) = \lambda_j\varphi_j(\mathbf{x}_k) \tag{3.3.3}$$

From 3.3.3, we can deduce,

$$\varphi_j(\mathbf{x}_k) = \lambda_j\varphi_j(\mathbf{x}_{k-1}) = \lambda_j[\lambda_j\varphi_j(\mathbf{x}_{k-2})] = \lambda_j^2[\lambda_j\varphi_j(\mathbf{x}_{k-3})] = \cdots = \lambda_j^k\varphi_j(\mathbf{x}_0) \tag{3.3.4}$$

Let us consider a vector of observable functions $\mathbf{g} := \begin{bmatrix} g_1 & g_2 & \dots & g_m \end{bmatrix}^T$ such that $\mathbf{g} : \mathcal{M} \to \mathbb{C}^m$ and $g_i \in \mathcal{F}$. If we consider that the eigenfunctions of the Koopman operator $\mathcal{K}$ spans each element

of g_i of $\mathbf{g}$, we can decompose $\mathbf{g}$ as the linear combination of the eigenfunctions of $\mathcal{K}$ (also Fourier series) as,

$$\mathbf{g}(\mathbf{x}_k) = \sum_{j=1}^{\infty} \varphi_j(\mathbf{x}_k)\boldsymbol{\eta}_j \tag{3.3.5}$$

with the vector co-efficients $\boldsymbol{\eta}_j$ called the Koopman modes (also dynamic modes) of $\mathbf{F}$ corresponding the observable $\mathbf{g}$.

Using 3.3.4 in 3.3.5,

$$\mathbf{g}(\mathbf{x}_k) = \sum_{j=1}^{\infty} \lambda_j^k \varphi_j(\mathbf{x}_0)\boldsymbol{\eta}_j \tag{3.3.6}$$

Therefore, the Koopman eigenvalues $\lambda_i \in \mathbb{C}$ characterize the temporal behavior of the corresponding Koopman mode $\boldsymbol{\eta}_j$: with the phase of λ_i corresponding to its frequency and the magnitude of λ_i corresponding to its growth rate.

For conservative dynamical systems or measure-preserving systems, for example Hamiltonian systems, the Koopman operator is unitary and it acts on the observable isometrically on the space $\mathcal{F} := \mathcal{L}^2(\mathcal{M})$ (square-integrable functions w.r.t. the conserved measure). Therefore, the Koopman eigenfunctions form an orthonormal basis for conservative systems [11] and we can compute Koopman modes $\boldsymbol{\eta}_j$ as,

$$\boldsymbol{\eta}_j = \begin{bmatrix} \langle \varphi_j, g_1 \rangle \\ \langle \varphi_j, g_2 \rangle \\ \dots \\ \langle \varphi_j, g_m \rangle \end{bmatrix} \tag{3.3.7}$$

3.3.1 Koopman Modes for a Linear System

Consider an n dimensional discrete-time autonomous linear system in a linear space $\mathcal{M}$ as given below,

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k) = \mathbf{A}\mathbf{x}_k \tag{3.3.8}$$

For a linear system, the eigenvalues of $\mathbf{A}$ are also the eigenvalues of the Koopman operator $\mathcal{K}$ [10]. Further, the eigenfunctions of $\mathcal{K}$ and the eigenvectors of $\mathbf{A}$ are related.

If $\mathbf{A}$ has a full set of n eigenvectors $\{\mathbf{v}_j\}_{j=1}^n$ with the corresponding eigenvalues $\{\lambda_j\}_{j=1}^n$, then

$$\mathbf{A}\mathbf{v}_j = \lambda_j \mathbf{v}_j \ , \ j = 1, 2, \dots, n \tag{3.3.9}$$

The eigenfunction φ_j of the Koopman operator can be found as [10],

$$\varphi_j(\mathbf{x}_k) = \langle \mathbf{x}_k, \mathbf{u}_j \rangle \ , \ j = 1, 2, \dots, n \tag{3.3.10}$$

where $\mathbf{u}_j$ is the eignvector of the adjoint $\mathbf{A}^*$ corresponding to the equation $\mathbf{A}^*\mathbf{u}_j = \bar{\lambda_j}\mathbf{u}_j$.

The equations 3.3.2,3.3.3, and 3.3.4 holds true for linear systems too, considering $\mathbf{g}(\mathbf{x}_k) = \mathbf{x_k}$ and with $j = 1, 2, \dots, n$ (since the full set of eigenvectors of $\mathbf{A}$ spans $\mathcal{M}$ and $\mathbf{g}$ here also evolves on $\mathcal{M}$).

Since $\mathbf{A}$ has a complete set of eigenvectors (forming an n-dimensional basis), we can write the evolution of the states as a linear combination of the eigenvectors as

$$\mathbf{x}_k = \mathbf{g}(\mathbf{x}_k) = \sum_{j=1}^{n} \alpha_j \mathbf{v}_j \ , \ \alpha_j \text{ is scalar} \tag{3.3.11}$$

Since $\mathbf{g}(\mathbf{x}_k)$ is observable, we can write it as a linear combination of the Koopman eigenfunctions and modes as,

$$\mathbf{g}(\mathbf{x}_k) = \sum_{j=1}^{n} \varphi_j(\mathbf{x}_k)\boldsymbol{\eta}_j \tag{3.3.12}$$

If we compare 3.3.12 with 3.3.6, it is clear that $\alpha_j = \varphi_j(\mathbf{x}_k)$ and $\boldsymbol{\eta}_j = \mathbf{v}_j$. Therefore, the Koopman modes $\boldsymbol{\eta}_j$ coincide with the eigenvectors $\mathbf{v}_j$ for linear systems.

Computing φ_j: Consider $\boldsymbol{\alpha} = \begin{bmatrix}\alpha_1 & \alpha_2 & \dots & \alpha_n\end{bmatrix}^T$, $\boldsymbol{\varphi} = \begin{bmatrix}\varphi_1 & \varphi_2 & \dots & \varphi_n\end{bmatrix}^T$, and the eigenvector matrix $\mathbf{V} = \begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_n\end{bmatrix}^T$, then 3.3.11/3.3.12 in matrix form is

$$\begin{aligned} \mathbf{x}_k &= \mathbf{V}\boldsymbol{\alpha} = \mathbf{V}\boldsymbol{\varphi}(\mathbf{x}_k) \\ \implies \boldsymbol{\alpha} &= \boldsymbol{\varphi}(\mathbf{x}_k) = \mathbf{V}^{-1}\mathbf{x}_k \end{aligned} \tag{3.3.13}$$

Using 3.3.6, the evolution of the states/observables for the linear system is given as,

$$\mathbf{g}(\mathbf{x}_k) = \mathbf{x}_k = \sum_{j=1}^{n} \lambda_j^k \varphi_j(\mathbf{x}_0)\mathbf{v}_j \tag{3.3.14}$$

Example: Consider the mass-spring-damper system represented by the following differential equation, $m\ddot{x} + c + kx = 0$. Considering $\mathbf{y} = \begin{bmatrix}y_1 & y_2\end{bmatrix}^T = \begin{bmatrix}x & \dot{x}\end{bmatrix}^T$, the equation of motion can be expressed in the state-space form of a system of coupled ODEs as,

$$\underbrace{\begin{bmatrix}\dot{y_1} \\ \dot{y_2}\end{bmatrix}}_{\dot{\mathbf{y}}} = \underbrace{\begin{bmatrix}0 & 1 \\ -k/m & -c/m\end{bmatrix}}_{\mathbf{A}} \begin{bmatrix}y_1 \\ y_2\end{bmatrix}$$

Consider $m = 1, k = 40$, and $c = 1.2$, and the initial conditions $x(0) = 0.2, \dot{x}(0) = 0.5$ and a sampling time of $0.001sec$. The discretized state space representation becomes,

$$\mathbf{y}_{k+1} = \mathbf{A}_d\mathbf{y}_k \text{ where, } \mathbf{A}_d = \begin{bmatrix}1 & 0.0019 \\ -0.04 & 0.9988\end{bmatrix} \tag{3.3.15}$$

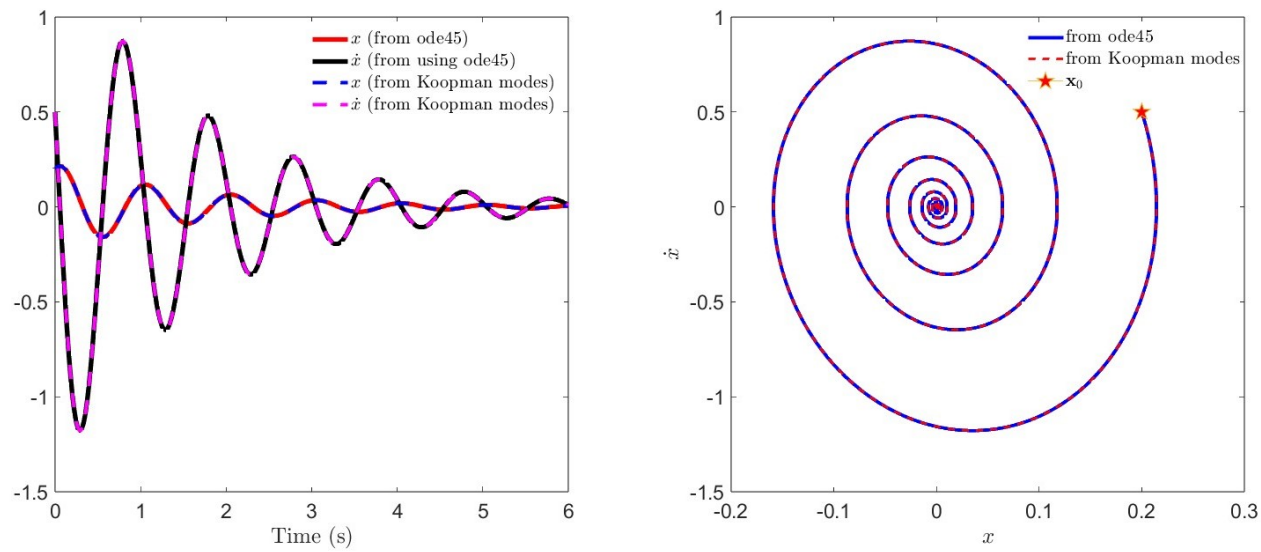

Figure 3.2: System states computed using the continuous time-model (numerically) and the Koopman modes (Code **??**)

3.3.2 Koopman Invariant Subspaces & Finite-Dimensional Linearization

Definition 3.3.1. *A **Koopman-invariant** subspace $\mathcal{F}_s$ can be defined as the span $\{g_j\}_{j=1}^m$ if for every $g_j \in \mathcal{F}_s$, $\mathcal{K}g_j \in \mathcal{F}_s$.*

Proposition 3.3.1. *A finite set of eigenfunctions of the Koopman operator will span an invariant subspace [11].*

Consider a discrete-time dynamical system evolving on a manifold $\mathcal{M}$ as in 3.2.3. The infinite-dimensional Koopman operator is defined on the space $\mathcal{F} \in \mathcal{H}$ where the observables g_j evolve in time. For practical applications, we are interested in finding a finite-dimensional subspace $\mathcal{F}_S$ of $\mathcal{F}$ which is Koopman invariant, i.e., for every $g_j(.) \in \mathcal{F}_S$ the corresponding $\mathcal{K}g_j(.) \in \mathcal{F}_S$. Consider a set of m linearly independent observable functions $\{g_j\}_{j=1}^m$ that spans $\mathcal{F}_S$ so we have a vector of observables $\mathbf{g} = \begin{bmatrix} g_1 & g_2 & \dots & g_m \end{bmatrix}^T$.

Since the subspace $\mathcal{F}_S$ is invariant under the operation of the Koopman operator, we can express $\mathcal{K}g_j$ as a linear combination of the observable functions g_j

$$\begin{aligned} \mathcal{K}g_j(\mathbf{x}_k) &= \alpha_{j_1} g_1(\mathbf{x}_k) + \alpha_{j_2} g_2(\mathbf{x}_k) + \cdots + \alpha_{j_m} g_m(\mathbf{x}_k) \\ &\implies \mathcal{K}g_j(\mathbf{x}_k) = \sum_{i=1}^{m} \alpha_{j_i} g_i(\mathbf{x}_k) \end{aligned} \tag{3.3.16}$$

Therefore, in matrix form,

$$\mathcal{K}\mathbf{g}(\mathbf{x}_k) = \underbrace{\begin{bmatrix} \alpha_{1_1} & \alpha_{1_2} & \dots & \alpha_{1_m} \\ \alpha_{2_1} & \alpha_{2_2} & \dots & \alpha_{2_m} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m_1} & \alpha_{m_2} & \dots & \alpha_{m_m} \end{bmatrix}}_{\mathbf{K}} \mathbf{g}(\mathbf{x}_k) \tag{3.3.17}$$

$$\implies \mathcal{K}\mathbf{g}(\mathbf{x}_k) = \mathbf{g}(\mathbf{x}_{k+1}) = \mathbf{K}\mathbf{g}(\mathbf{x}_k) \tag{3.3.18}$$

Here, $\mathbf{K}$ is a matrix operator, representing $\mathcal{K}$, which linearly transforms the vector observable $\mathbf{g}$ on the invariant subspace $\mathcal{F}_S$. Therefore, any eigenvalue λ_j of $\mathbf{K}$ is also an eigenvalue of $\mathcal{K}$. Let $\mathbf{v}_j$ be an eigenvector of $\mathbf{K}$ and the eigenvector matrix (with m eigenvectors which may contain generalized ones) is $\mathbf{V} = \begin{bmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \dots & \mathbf{v}_m\end{bmatrix}$

Consider a set of m eigenfunctions $\boldsymbol{\varphi} = \begin{bmatrix}\varphi_1 & \varphi_2 & \dots & \varphi_m\end{bmatrix}^T$ of $\mathcal{K}$ on $\mathcal{F}_S$. Since $\{g_j\}_{j=1}^m$ span the subspace $\mathcal{F}_S$, we can write $\boldsymbol{\varphi}$ as linear combinations of $\{g_j\}_{j=1}^m$ like a Fourier series with the vector coefficients being the Koopman modes $\boldsymbol{\eta}_j$ as,

$$\boldsymbol{\varphi}(\mathbf{x}_k) = \sum_{j=1}^{m} g_j(\mathbf{x}_k)\boldsymbol{\eta}_j \tag{3.3.19}$$

Further, the eigenfunctions also evolve linearly under the operation of $\mathcal{K}$ as,

$$\mathcal{K}\boldsymbol{\varphi}(\mathbf{x}_k) = \boldsymbol{\varphi}(\mathbf{x}_{k+1}) = \mathbf{K}\boldsymbol{\varphi}(\mathbf{x}_k) \tag{3.3.20}$$

Therefore, we can write the eigenfunctions as linear combinations of the eigenvectors of $\mathbf{K}$ as,

$$\boldsymbol{\varphi}(\mathbf{x}_k) = \sum_{j=1}^{m} \alpha_j \mathbf{v}_j = \mathbf{V}\boldsymbol{\alpha} \tag{3.3.21}$$

Where, $\boldsymbol{\alpha} = \begin{bmatrix}\alpha_1 & \alpha_2 & \dots & \alpha_n\end{bmatrix}^T$

From 3.3.19 and 3.3.21, $\alpha_j = g_j(\mathbf{x}_k)$, therefore the Koopman eigenfunctions can be computed as,

$$\boldsymbol{\varphi}(\mathbf{x}_k) = \sum_{j=1}^{m} g_j(\mathbf{x}_k)\mathbf{v}_j = \mathbf{V}\mathbf{g}(\mathbf{x}_k) \tag{3.3.22}$$

To verify that 3.3.22 represents the eigenfunctions of $\mathcal{K}$,

$$\mathcal{K}\boldsymbol{\varphi}(\mathbf{x}_k) = \mathbf{K}\boldsymbol{\varphi}(\mathbf{x}_k) = \mathbf{K}\mathbf{V}\mathbf{g}(\mathbf{x}_k) = \boldsymbol{\Lambda}\mathbf{V}\mathbf{g}(\mathbf{x}_k) = \boldsymbol{\Lambda}\boldsymbol{\varphi}(\mathbf{x}_k) \tag{3.3.23}$$

Where, $\boldsymbol{\Lambda}$ is the diagonal matrix of the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$.

The idea of subspace-invariance of the Koopman operator is helpful as it allows us to do our computations over a finite-dimensional subspace. Even if the subspace $\mathcal{F}_s \subseteq \mathcal{F}$ is not invariant under the Koopman operator, we can still use the notion of subspace invariance. In this case, we can consider the subspace $\mathcal{F}_s$ is invariant under the modified operator $\mathcal{P}_{\mathcal{F}_s}\mathcal{K} : \mathcal{F} \to \mathcal{F}$. Here, $\mathcal{P}_{\mathcal{F}_s} : \mathcal{F} \to \mathcal{F}$ is the orthogonal projection operator on $\mathcal{F}_s$. In this case,

$$\mathcal{K}\boldsymbol{\varphi}(\mathbf{x}_k) \approx \mathcal{P}_{\mathcal{F}_s}\mathcal{K}\boldsymbol{\varphi}(\mathbf{x}_k) = \hat{\mathbf{K}}\boldsymbol{\varphi}(\mathbf{x}_k) \tag{3.3.24}$$

Discovering the Koopman eigenfunction coordinates is, therefore, a central challenge, as they provide intrinsic coordinates along which the dynamics behave linearly. In practice, it is more likely that we will identify an approximately invariant subspace, given by a set of functions $\{g_j\}_{j=1}^m$, where each g_j is approximated by a finite sum of eigenfunctions (as in (3.3.22)) [11]).

3.4 Introduction to Dynamic Mode Decomposition (DMD)

Dynamic Mode Decomposition (DMD) is a standard algorithm to approximately estimate the Koopman operator from data. It is based on the singular value decomposition (SVD) providing scalable dimensionality reduction for high dimensional data. The DMD algorithm looks for a best-fit linear operator $\mathbf{A}$ that approximately advances the states of a system forward in time. Thus, $\mathbf{A}$ is an approximation of the Koopman operator $\mathcal{K}$ restricted to a measurement subspace spanned by direct measurements of the state ($\mathbf{g}(\mathbf{x}_k) = \mathbf{x}_k \implies \mathcal{K} * \mathbf{x}_k = \mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k$).

DMD is inherently a data-driven algorithm and the matrix $\mathbf{A}$ is approximated from a collection of snapshot pairs of the system. Consider a system with $\mathbf{x} \in \mathbb{R}^n$ with the state measurements in the time interval $\mathcal{I} := [t_0, t]$. Let the measurements are taken in discrete-intervals $k+1$ steps in $\mathcal{I}$ as $\{t_0, t_1, t_2, \ldots, t_k\}$ with sampling time $\Delta t = t_i - t_{i-1}$ so that the discrete time evolution of the state vector is given as $\{\mathbf{x}_{t_0}, \mathbf{x}_{t_1}, \ldots, \mathbf{x}_{t_k}\} := \{\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2 \ldots, \mathbf{x}_k\}$. Therefore, a snapshot pair of the state measurements become $\{\mathbf{x}_p, \mathbf{x}_{p+1}\}_{p=0}^{k}$. in the original DMD formulation by Schmid required data from a single trajectory with uniform sampling in time. However, the DMD we present here is the *exact DMD* algorithm of Tu et al. In exact DMD, the time need not be sequential or evenly spaced, but the snapshot pair should be sequential.

Consider the dynamics of the system in discrete time with one snapshot pair,

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k \tag{3.4.1}$$

We arrange the snapshot pairs into two $n \times k$ data matrices,

$$\begin{aligned} \mathbf{X} &= \begin{bmatrix} | & | & \cdots & | \\ \mathbf{x}_0 & \mathbf{x}_1 & \cdots & \mathbf{x}_{k-1} \\ | & | & \cdots & | \end{bmatrix} \\ \mathbf{X}' &= \begin{bmatrix} | & | & \cdots & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_k \\ | & | & \cdots & | \end{bmatrix} \end{aligned} \tag{3.4.2}$$

Therefore, (3.4.1) can be written in matrix form as,

$$\mathbf{X}' \approx \mathbf{A}\mathbf{X} \tag{3.4.3}$$

The best-fit matrix $\mathbf{A}$ establishes a linear dynamical system that linearly advances snapshot measurement in forward time and can be obtained by formulating the optimization problem as follows,

$$\mathbf{A} = \underset{\mathbf{A}}{\operatorname{argmin}} ||\mathbf{X}' - \mathbf{A}\mathbf{X}||_F = \mathbf{X}'\mathbf{X}^{\dagger} \tag{3.4.4}$$

where $\mathbf{X}^{\dagger}$ is the pseudo-inverse of $\mathbf{X}$ and $||.||_F$ is the Frobenius norm.

$\mathbf{X}^\dagger$ can be computed using the SVD of $\mathbf{X}$

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^* \tag{3.4.5}$$

such that,

$$\mathbf{X}^\dagger = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^* \tag{3.4.6}$$

The columns of $\mathbf{U}$ are called **POD** (proper orthogonal decomposition) modes.

Since $\mathbf{A}$ is an approximate representation of the Koopman operator restricted to a finite-dimensional subspace of linear measurements, we are interested in computing the eigenvalues and eigenvectors of $\mathbf{A}$. For $k > n$ and n being reasonably small, we can conveniently obtain the spectrum of $\mathbf{A}$. However, it is not convenient to compute the spectrum when the dimension of $\mathbf{A}$ is enormous.

3.4.1 DMD with leading spectral decomposition

DMD was introduced for high-dimensional systems, such as fluid dynamics, where the state dimension n may be huge. Therefore, the DMD algorithm seeks the leading spectral decomposition without explicitly constructing it. In such cases, $k \ll n$. Thus, the effective rank of the $\mathbf{A}$ can be at most k, and it can be even lower. If we have $\mathbf{A}$ with rank $r < k$, we truncate the SVD of $\mathbf{X}$ into its first r POD modes as,

$$\mathbf{X} = \mathbf{U}_r\mathbf{\Sigma}_r\mathbf{V}_r^* \tag{3.4.7}$$

In this case, the $n-$dimensional space of the states is related to the $r-$dimensional space of POD modes as,

$$\mathbf{X} = \mathbf{U}_r\mathbf{X}_r \text{ or } \mathbf{X}_r = \mathbf{U}_r^*\mathbf{X} \tag{3.4.8}$$

Therefore, $\mathbf{A}$ (with dimension $> r$) can be projected onto the first $r-$POD modes as,

$$\begin{aligned} \mathbf{A}_r &= \mathbf{U}_r^*\mathbf{A}\mathbf{U}_r \\ &= \mathbf{U}_r^*\mathbf{X}'\mathbf{X}^\dagger\mathbf{U}_r \\ &= \mathbf{U}_r^*\mathbf{X}'\mathbf{V}_r\mathbf{\Sigma}_r^{-1}\mathbf{U}_r^*\mathbf{U}_r \\ \therefore \mathbf{A}_r &= \mathbf{U}_r^*\mathbf{X}'\mathbf{V}_r\mathbf{\Sigma}_r^{-1} \end{aligned} \tag{3.4.9}$$

The spectrum and eigenvectors of the reduced $\mathbf{A}_r$ can be computed from,

$$(\mathbf{A}_r - \mathbf{\Lambda}\mathbf{I})\mathbf{W} = 0 \tag{3.4.10}$$

where $\mathbf{W}$ is the eigenvector matrix and $\mathbf{\Lambda}$ is the diagonal matrix with the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_r$ at the diagonal. $\mathbf{\Lambda}$ is also the DMD eigenvalues of the high-dimensional $\mathbf{A}$ matrix.

The DMD modes are the eigenvectors of $\mathbf{A}$. Tu et al. proved that $\mathbf{\Phi}$ is eigenvector matrix of $\mathbf{A}$ reconstructed from $\mathbf{W}$ given as,

$$\mathbf{\Phi} = \mathbf{X}'\mathbf{V}_r\mathbf{\Sigma}^{-1}\mathbf{W} \tag{3.4.11}$$

Once we obtain the DMD modes, the system states can be expressed in terms of DMD expansion (in discrete time) as,

$$\mathbf{x}_k = \sum_{i=1}^{r} \phi_i \lambda_i^{k-1} b_j = \mathbf{\Phi}\mathbf{\Lambda}^{k-1}\mathbf{b} \tag{3.4.12}$$

where,

$$\mathbf{b} = \mathbf{\Phi}^{\dagger}\mathbf{x}_0 \tag{3.4.13}$$

The DMD expansion in (3.4.12) is related to the Koopman mode decomposition. Therefore,using DMD the Koopman mode decomposition can be written as,

$$\mathbf{x}_k = \sum_{i=1}^{r} \lambda_i^{k-1} \varphi_i(\mathbf{x}_1)\mathbf{v}_i \tag{3.4.14}$$

where, $\phi_i = \mathbf{v}_i$ (columns of $\mathbf{\Phi}$) and $\varphi_i(\mathbf{x}_1)$ is the *i-th* element of $\mathbf{b} = \mathbf{\Phi}^{\dagger}\mathbf{x}_1$.

In continuous time, the eigenvalues ω are related to the discrete-time eigenvalues λ of $\mathbf{A}$ as,

$$\lambda = e^{\omega \Delta t} \implies \omega = \frac{ln(\lambda)}{\Delta t} \tag{3.4.15}$$

Therefore, in continuous time, the DMD expansion will give rise to the data matrices, given as,

$$\mathbf{X} \approx \sum_{i=1}^{r} \phi_i e^{\omega_i t} b_j = \mathbf{\Phi} e^{\mathbf{\Omega} t}\mathbf{b} \tag{3.4.16}$$

where $\mathbf{\Omega}$ is the diagonal matrix with the continuous-time eigenvalues at the diagonal.

Example 3.4.1. *Consider the spatio-temporal data and its DMD approximation as shown in Figure 3.4. This is case where $n \gg k$.*

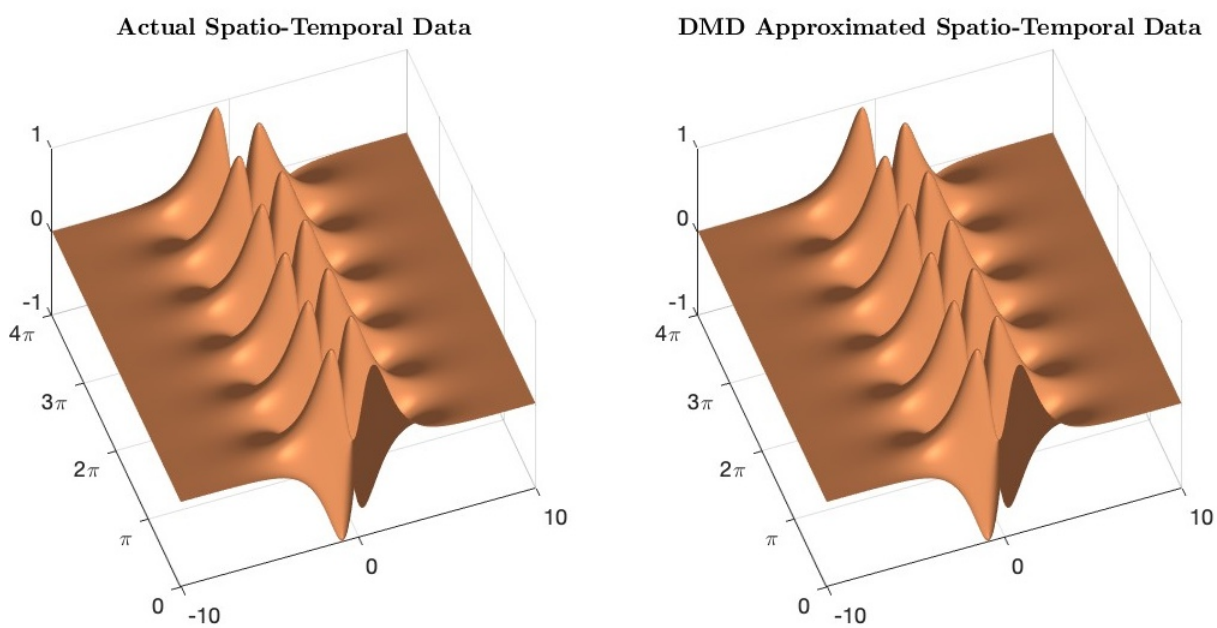

Figure 3.3: DMD approximation of a spatio-temporal data

Example 3.4.2. *Consider the autonomous system given as (this is a case where $k > n$).*

$$\begin{aligned} \dot{x}_1 &= -x_2 + x_1 x_2^2 \\ \dot{x}_2 &= x_1 - x_1^2 x_2 \end{aligned} \tag{3.4.17}$$

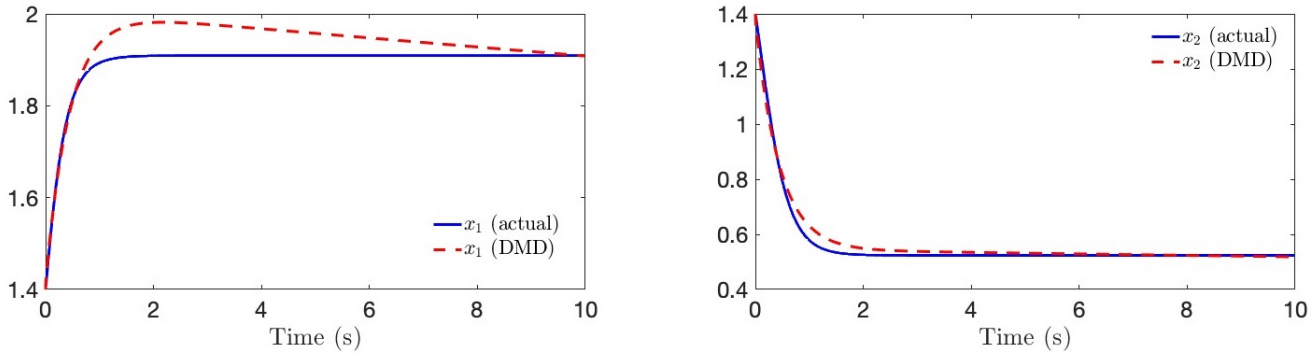

Figure 3.4: DMD approximation for $x_1(0) = x_2(0) = 1.4$

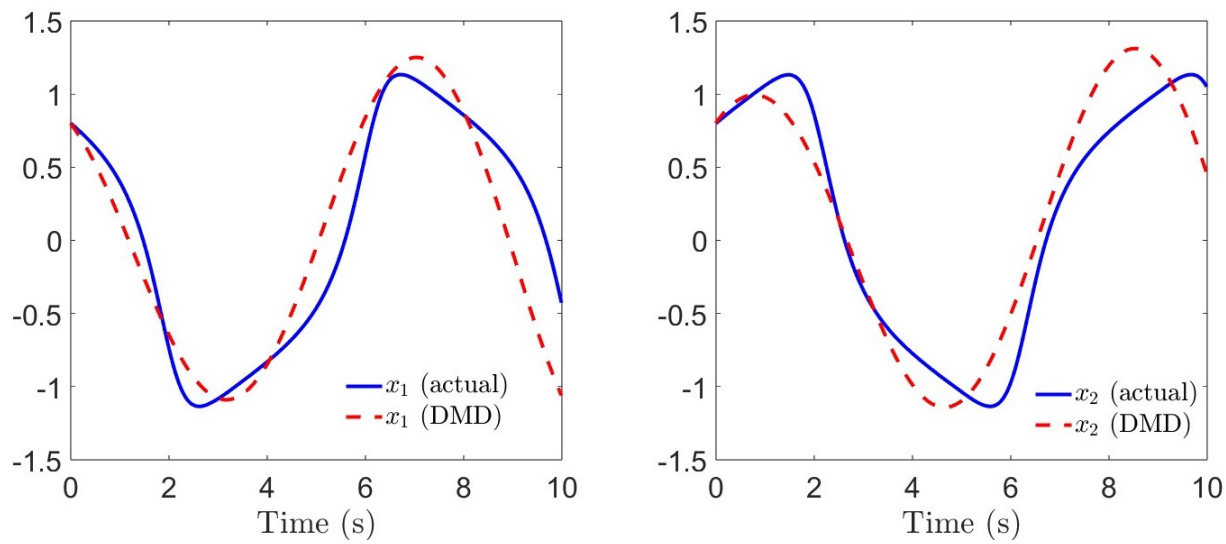

Figure 3.5: DMD approximation for $x_1(0) = x_2(0) = 0.8$

Chapter 4

Bilinear Model Learning

4.1 Bilinear Models Approximating Koopman Generator

Consider a control-affine non-linear system,

$$\begin{aligned} \dot{\mathbf{x}} = \mathbf{f}_0(\mathbf{x}) + \sum_{i=1}^{m} u_i(t)\mathbf{f}_i(\mathbf{x}) \\ \mathbf{y} = \mathbf{g}(\mathbf{x}) \end{aligned} \tag{4.1.1}$$

where, $\mathbf{x} \in \mathcal{X} \subseteq \mathbb{R}^n$, $u_i \in \mathbb{R}$ (for $i = 1, 2, \ldots, m$), and $\mathbf{y} \in \mathbb{R}^p$ ($p < n$). Control inputs enter the system only through control vector fields $\mathbf{f}_i$. Consider a set of observable functions $\mathbf{\Psi} = \{\Psi_1, \Psi_2, \ldots, \Psi_N\}$ such that $\mathbf{g} = \mathbf{C}\mathbf{\Psi}$ and $\Psi_j : \mathcal{X} \to \mathbb{R}$. Clearly. $\mathbf{g}$ might not be injective for some relevant observables, thereby providing incomplete information about the system states related to those observables.

Therefore, the evolution of the observables in time is given by the Koopman generator as,

$$\frac{\partial \Psi_j}{\partial t} = L_u \Psi_j = L_{\mathbf{f}_0}\Psi_j + \sum_{i=1}^{m} u_i(t) L_{\mathbf{f}_i}\Psi_j \tag{4.1.2}$$

where the Koopman generator $L_u = \lim_{t\to 0} \frac{\mathcal{K}_u^t - I}{t}$ such that $\mathcal{K}_u^t \Psi_j(\mathbf{x}_0) = \Psi_j(\mathbf{x})$.

Considering finite-dimensional approximation of the Koopman generator with the N observables results in the following bilinear expression,

$$\frac{\partial \Psi_j}{\partial t} = L_u \Psi_j \approx (\mathbf{l}_{\mathbf{f}_0})_j \Psi_j + \sum_{i=1}^{m} u_i(t)(\mathbf{l}_{\mathbf{f}_i})_j \Psi_j \tag{4.1.3}$$

Considering $\mathbf{z} = \mathbf{\Psi}(\mathbf{x})$ the lifted bilinear model based on the approximated Koopman generator is defined as

$$\begin{aligned} \dot{\mathbf{z}} = \mathbf{V}_0\mathbf{z} + \sum_{i=1}^{m} u_i(t)\mathbf{V}_i\mathbf{z} = \left(\mathbf{I} + \textstyle\sum_{i=0}^{m} u_i(t)\mathbf{V}_i\right)\mathbf{z} \\ \mathbf{y} = \mathbf{C}\mathbf{z} \end{aligned} \tag{4.1.4}$$

where,

$$\mathbf{V}_0 = \begin{bmatrix} | & | & \dots & | \\ (\mathbf{l}_{\mathbf{f}_0})_1 & (\mathbf{l}_{\mathbf{f}_0})_2 & \dots & (\mathbf{l}_{\mathbf{f}_0})_N \\ | & | & \dots & | \end{bmatrix}, \text{ and } \mathbf{V}_i = \begin{bmatrix} | & | & \dots & | \\ (\mathbf{l}_{\mathbf{f}_i})_1 & (\mathbf{l}_{\mathbf{f}_i})_2 & \dots & (\mathbf{l}_{\mathbf{f}_i})_N \\ | & | & \dots & | \end{bmatrix} \tag{4.1.5}$$

4.1.1 Lifted LTI Model vs Bilinear Model

A lifted LTI model with $\mathbf{z} = \mathbf{\Psi}(\mathbf{x})$ is defined as,

$$\begin{aligned} \dot{\mathbf{z}} &= \mathbf{A}\mathbf{z} + \mathbf{B}\mathbf{u} \\ \mathbf{y} &= \mathbf{C}\mathbf{z} \end{aligned} \tag{4.1.6}$$

The lifted LTI model is a special case of the bilinear model in (4.1.4) using an augmented set of observables $\mathbf{\Psi} = \{1, \Psi_1, \Psi_2, \dots, \Psi_N\}$, given as,

$$\begin{bmatrix} 0 \\ \dot{\mathbf{z}} \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & \mathbf{A} \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{z} \end{bmatrix} + \sum_{i=1}^{m} u_i \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{b}_i & \mathbf{0} \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{z} \end{bmatrix}, \text{ with } \mathbf{B} = \begin{bmatrix} | & \dots & | \\ \mathbf{b}_1 & \dots & \mathbf{b}_m \\ | & \dots & | \end{bmatrix} \tag{4.1.7}$$

4.2 Approximation of the Koopman Generator using HMM

Consider we have a collection of finely sampled time histories of the input and the observations $\{\mathbf{u}_k\}_{k=0}^{K-1}$ and $\{\mathbf{y}_k\}_{k=0}^{K}$. We have M such measurements or trajectories. We seek to estimate the Koopman generators $\{\mathbf{V}_i\}_i^k$ together with an appropriate collection $\mathbf{\Psi}$ (not known beforehand) that fits the observed data as closely as possible. Forcing one of the states to be constant (i.e., $\mathbf{\Psi} = \{1, \Psi_1, \Psi_2, \dots, \Psi_N\}$), we can express the bilinear model in (4.1.4) in a form similar to (4.1.7) with a constant shift in the output such that,

$$\mathbf{C} = \begin{bmatrix} \mathbf{c}_0 & \mathbf{I}_{p\times p} & \mathbf{0} \end{bmatrix} \tag{4.2.1}$$

With finer sampling of data (fixed at Δt)), we can think of a hidden Markov model (HMM) representing the dynamics of the collection of observables using explicit Euler discretization of (4.1.4) and the above-mentioned constant embedding. We consider combined process noise + modeling error as Gaussian random vector $\mathbf{w}_k \approx \mathcal{N}(\mathbf{0}, \mathbf{\Sigma_w})$ and combined measurement noise + modeling error as Gaussian random vector $\mathbf{d}_k \approx \mathcal{N}(\mathbf{0}, \mathbf{\Sigma_d})$ for each trajectory of collected data. Therefore, the bilinear hidden Markov model (HMM) is represented as,

$$\begin{aligned} \begin{bmatrix} 1 \\ \mathbf{z}_{k+1} \end{bmatrix} &= \left(\mathbf{I} + \Delta t \textstyle\sum_{i=0}^{m} u_i(t)\mathbf{V}_i\right) \begin{bmatrix} 1 \\ \mathbf{z}_k \end{bmatrix} + \begin{bmatrix} 0 \\ \mathbf{w}_k \end{bmatrix} \\ \mathbf{y}_k &= \mathbf{c}_0 + \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{z}_k + \mathbf{d}_k \end{aligned} \tag{4.2.2}$$

The parameters of the HMM we seek to determine the following parameters from data,

$$\Theta = (\{\mathbf{V}_i\}_{i=0}^{m}, \mathbf{c}_0, \mathbf{\Sigma_w}, \mathbf{\Sigma_d}, \boldsymbol{\mu}_0, \mathbf{\Sigma_0}) \tag{4.2.3}$$

where the initial condition is also considered a Gaussian distribution such that $\mathbf{z}_0 \approx \mathcal{N}(\boldsymbol{\mu}_0, \mathbf{\Sigma_0})$.

(4.2.2) can be equivalently expressed as,

$$\begin{bmatrix} 1 \\ \mathbf{z}_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{b} & \mathbf{A} \end{bmatrix} \begin{bmatrix} 1 \\ \mathbf{z}_k \end{bmatrix} + \begin{bmatrix} 0 \\ \mathbf{w}_k \end{bmatrix} \\ \mathbf{y}_k = \mathbf{c}_0 + \begin{bmatrix} \mathbf{I} & \mathbf{0} \end{bmatrix} \mathbf{z}_k + \mathbf{d}_k \tag{4.2.4}$$

with

$$\begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{b}_k & \mathbf{A}_k \end{bmatrix} = \left(\mathbf{I} + \Delta t \sum_{i=0}^{m} u_i(t) \mathbf{V}_i\right) \tag{4.2.5}$$

such that the latent dynamics of the HMM is,

$$\mathbf{z}_{k+1} = \mathbf{A}_k \mathbf{z}_k + \mathbf{b}_k + \mathbf{w}_k \tag{4.2.6}$$

$$\boldsymbol{\mu}_k = \mathbb{E}[\mathbf{z}_k | \mathbf{Y} = [\mathbf{y}_0, \ldots, \mathbf{y}_k]] \tag{4.2.7}$$

Chapter 5

Appendix

5.1 MATLAB Codes

Please contact the author (rajkumar.36@osu.edu) if you would like to have some of the codes used in this document.

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